

Comparative Analysis of Piezoelectric Materials in a Sandwich Aircraft Wing

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Abstract—In the modern era of aerospace engineering, with an increasing demand for lightweight and multifunctional structures, vibration energy harvesting is a vital aspect for powering embedded sensors and Structural Health Monitoring (SHM) systems. The wing of an aircraft undergoes mechanical vibrations because of aerodynamic forces, turbulence, and engine excitation. Thus, harvesting otherwise dissipated vibrational energy converts operational vibrations into usable electrical energy, thereby minimizing the use of traditional batteries. This paper discusses a detailed comparative numerical study on the integration of piezoelectric layers in a three-dimensional sandwich aircraft wing. A half-wing structure with a Clark-Y airfoil section was modeled in CATIA V5 and analyzed numerically in ANSYS Mechanical 2025 R2. Five commercially available Lead Zirconate Titanate ceramic materials — PZT-4, PZT-5A, PZT-5H, PZT-8, and PZT-26 were integrated into the sandwich intermediate layer and subjected to similar static structural, free vibration modal, and coupled-field harmonic analysis under steady and swept frequency aerodynamic pressure. From the numerical results, it is evident that PZT-26 has the least static deflection (0.1387 mm) and the highest fundamental frequency (54.193 Hz), thereby exhibiting better stiffness both structurally and dynamically. At the same time, PZT-8 produced the highest peak electrical voltage (6275.1 mV) under harmonic excitation. PZT-26 showed the least static deflection and the highest fundamental frequency, suggesting the best structural and dynamic behavior among all the materials investigated.

Keywords—Piezoelectric Materials, Lead Zirconate Titanate (PZT), Vibration Energy Harvesting, Finite Element Analysis (FEA), Modal Analysis, Coupled-Field Harmonic Analysis, Structural Health Monitoring (SHM), Sandwich Aircraft Wing.

I. INTRODUCTION

The advancement of aerospace technology has increased the requirement for light, efficient, and intelligent structural systems which improve the performance of the aircraft while reducing maintenance problems [1]. The aircraft structures

today need to not only endure aerodynamic forces, bending forces, and torsional vibrations in the air, but they also have to serve the role of SHM and power generation [6].

Of all techniques used for the collection of ambient energy, vibration energy collection through the use of piezoelectric materials is highly beneficial for aerospace applications [7]. There exist constant ambient vibrations in aircraft wings due to aeroelastic flutter, turbulence, and excitation from engines [2], [5]. Through the integration of piezoelectric sensors in structural elements, the vibrations can be converted to electricity through the principle of direct piezoelectric effect [10]. The collected electricity can be used to power sensor nodes, electronic circuits, and transducers for monitoring, thus making it unnecessary to change batteries [1], [6].

Sandwich structure arrangements that feature hard and thin facesheets with a light core in between have very high flexural stiffness and weight-to-strength ratios [4], [11]. Incorporation of piezoelectric ceramics in sandwich aircraft wing skins produces a smart material that can perform the tasks of carrying loads and generating electricity [5], [9]. Although some research studies have mainly concentrated on the performance characteristics of the piezoelectric sandwich plates, piezoelectric energy harvesters, or the particular structural designs, it is not yet well understood how different PZT ceramic materials behave structurally and electrically in the context of a comparative analysis in a single sandwich aircraft wing design. This can give some useful insights to make a proper choice of the material. This study offers a comparison of finite element analysis of five grades of commercially available Lead Zirconate Titanate (PZT) ceramic materials: PZT-4, PZT-5A, PZT-5H, PZT-8, and PZT-26 in a smart sandwich aircraft wing skin.

II. THEORETICAL BACKGROUND AND GOVERNING EQUATIONS

A. Fundamentals of Piezoelectricity

Piezoelectricity refers to the electromechanical coupling exhibited by non-centrosymmetric crystals between the

electrical domain and the mechanical domain [7], [10]. Direct piezoelectricity is caused by the generation of electrical charge displacement by mechanical strain, which is used in energy harvesting devices and sensors [1]. On the other hand, inverse piezoelectricity causes mechanical strain due to the application of electric fields for precise actuation [2].

The linear constitutive equations describing piezoelectric materials under coupled electromechanical state are written as [10]:

$$S = s^E T + d^T E \quad (1)$$

$$D = d T + \varepsilon^T E \quad (2)$$

Where S is mechanical strain, s^E is elastic compliance under constant electric field, T is mechanical stress, d is the piezoelectric strain coefficient matrix, E is the electric field, D is electric displacement vector, and ε^T is the dielectric permittivity under constant stress [7], [10].

B. Mechanics of Smart Sandwich Wings

A wing in an airplane is mechanically equivalent to a cantilever beam that has one end fixed and the other end free [2]. Aerodynamic distribution of forces on a wing causes maximum bending moment and transverse shear stress at the root of the wing. In a sandwich structure, the upper and lower face sheets resist bending tension and compression forces, while the intermediate core sheet carries the transverse shear force and keeps the face sheets apart [4]. By incorporating piezoelectric layers in the core, mechanical strain can be transferred effectively into the piezoelectric layer [5], [9].

C. Governing Equations of Structural and Dynamic Behavior

The deformations due to structural loading and von Mises stresses are related using linear elasticity theory equations [2]:

$$\sigma = E \varepsilon \quad (3)$$

The aerodynamic pressure q on the wing skin can be determined using the flight velocity V and the air density ρ according to dynamic pressure:

$$q = 0.5 \rho V^2 \quad (4)$$

Free vibration behavior and natural frequencies f_n are analyzed by using the structure undamped eigenvalue analysis approach [8]:

$$(K - \omega^2 M) \varphi = 0 \quad (5)$$

where K represents the global stiffness matrix, M represents the global mass matrix, $\omega = 2\pi f_n$ is the natural frequency in circular form, and φ represents the mode shape vector [8].

Under harmonic loading conditions, the dynamics of the mechanical structure is governed by the equation [2], [10]:

$$M u'' + C u' + K u = F(t) \quad (6)$$

III. MATERIALS SELECTION AND CAD MODELING



Fig. 1: Overall research methodology workflow adopted for the numerical investigation.

A. Material Selection

The basic parts of the wing structure including skin, spars, and ribs were made of aluminum alloy [4]. Five types of commercially available PZT ceramic materials named as PZT-4, PZT-5A, PZT-5H, PZT-8, and PZT-26 were chosen in order to study the effect of piezoelectric material selection on structural, dynamical, and electromechanical behavior of the smart wing. The main properties and applications of chosen materials are given in Table I [1], [3], [7].

TABLE I: COMPARISON OF PIEZOELECTRIC MATERIALS EVALUATED

Material	Class	Main Characteristics	Primary Application
PZT-4	Hard Ceramic	High stiffness, low dielectric loss	High-power transducers & vibration control
PZT-5A	Soft Ceramic	High sensitivity, balanced properties	Sensors, actuators, general harvesting
PZT-5H	Soft Ceramic	Very high d33 & coupling factor	Precision actuators, low-freq harvesters
PZT-8	Hard Ceramic	High Qm, low dielectric loss	High-power ultrasonic & aerospace transducers
PZT-26	Modified	High dynamic stability & stiffness	Smart structural monitoring & harvesting

B. Geometric CAD Modeling

A three-dimensional computer-aided design (CAD) model of a rectangular half-wing was created using CATIA V5 software based on the Clark-Y airfoil section. The design includes two longitudinal hollow square spars with 500 mm span each, six evenly placed ribs, and three-layer segmented skin of the wing.

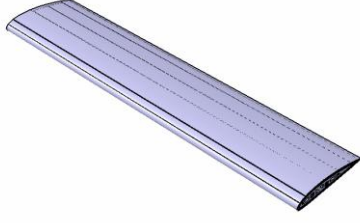


Fig. 2. Complete 3D CAD model of the rectangular half-wing developed in CATIA V5.

TABLE II: GEOMETRICAL SPECIFICATIONS OF AIRCRAFT WING

Parameter	Specification Value
Wing Configuration	Rectangular Half-Wing
Airfoil Profile	Clark-Y
Half Span (b)	500.00 mm
Root & Tip Chord (c)	115.84 mm
Maximum Airfoil Thickness	13.55 mm (11.7% c)
Face Skin Thickness (Upper/Lower)	1.00 mm / 1.00 mm
Intermediate Layer Thickness	1.00 mm
Total Sandwich Thickness	3.00 mm
Spars & Ribs Configuration	2 Square Hollow Spars, 6 Ribs

TABLE III: STRUCTURAL SPAR DIMENSIONS

Parameter	Dimension / Property
Number of Spars	2
Cross Section	Hollow Square
Outer Dimensions	5.00 mm $\times$ 5.00 mm
Wall Thickness	0.50 mm
Front Spar Position	25.482 mm from Leading Edge
Rear Spar Position	50.481 mm from Leading Edge

TABLE IV: STRUCTURAL RIB DIMENSIONS

Parameter	Dimension / Property
Number of Ribs	6
Rib Thickness	3.00 mm
Rib Spacing Pitch	99.40 mm
Internal Features	Lightening Holes & Spar Pockets

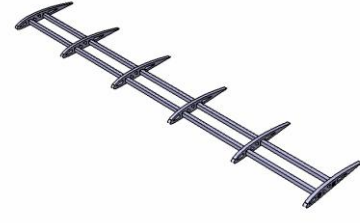


Fig. 3. Internal structural arrangement of the aircraft wing showing spars and ribs.

In addition to the inner and outer skins, the intermediate core consists of eight separate segments which include two aluminum edge segments and 6 ceramic piezoelectric core segments that are arranged in areas where maximum bending strain occurs [6], [9].

TABLE V: SANDWICH SKIN PANEL SPECIFICATIONS

Panel Layer Component	Material & Segment Layout
Top Outer Skin	Aluminum Alloy (2 Panels, 1.0 mm thick)
Intermediate Core Layer	8 Chordwise Segments (2 Al, 6 PZT)
Bottom Outer Skin	Aluminum Alloy (2 Panels, 1.0 mm thick)
Active Piezoelectric Zone	Central 6 PZT Segments (1.0 mm thick)

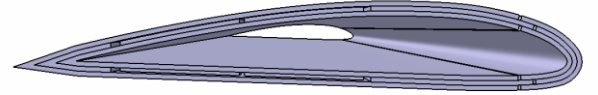


Fig. 4. Segmented aluminum and piezoelectric intermediate layer arrangement within the sandwich skin.

IV. FINITE ELEMENT ANALYSIS SETUP

A. Engineering Data and Finite Element Meshing

Finite Element Modeling was performed using ANSYS Mechanical 2025 R2 [2]. Mechanical properties of structural Aluminum Alloy material were defined as shown in Table VI. Refinement of mesh in skin and piezoelectric interface layers was done owing to the presence of steep strain gradients [8].

TABLE VI: MECHANICAL PROPERTIES OF ALUMINUM ALLOY

Engineering Property	Value
Density (ρ)	2.77×10^{-6} kg/mm ³ (2770 kg/m ³)

Young's Modulus (E)	71,000.00 MPa (71 GPa)
Poisson's Ratio (ν)	0.33
Yield Strength (σ_y)	280.00 MPa
Ultimate Tensile Strength (σ_{uts})	310.00 MPa

TABLE VII: FINITE ELEMENT MESH STATISTICS

Mesh Statistic Parameter	Value
Total Number of Nodes	608,002
Total Number of Elements	85,091
Element Type	3D Solid Hexahedral & Tetrahedral
Contact Condition	Bonded Interfaces Across All Layers

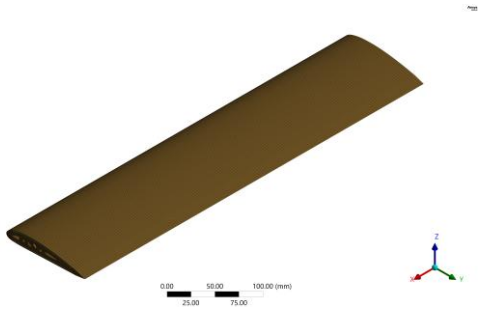


Fig. 5. Finite element mesh of the smart wing model in ANSYS Mechanical.

B. Loading and Boundary Conditions

A Fixed Support condition was imposed at the wing root station ($x = 0$ mm) to indicate the presence of rigidity between the wings and fuselage [2]. The dynamic pressure $q = 250$ Pa (2.5×10^{-4} MPa), resulting from a flying speed of $V = 20.2032$ m/s at standard atmospheric conditions at sea level with density $\rho = 1.225$ kg/m³, was set on the underside skin surface.

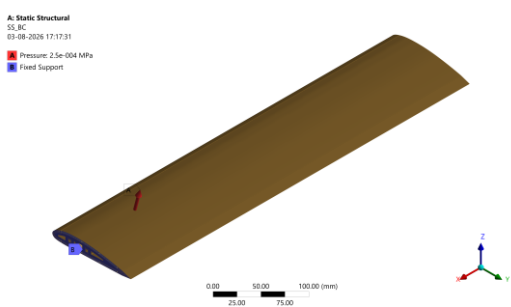


Fig. 6. Static structural loading and boundary conditions applied to the aircraft wing model.

For the coupled field harmonic analysis, the electrical boundary conditions were determined using 0 V Ground potential applied to the bottom surface of the PZT segments, while the top surface was coupled to determine the electrical potential generated as a result of the various excitation frequencies between 0 Hz and 1160.3 Hz [6], [9].

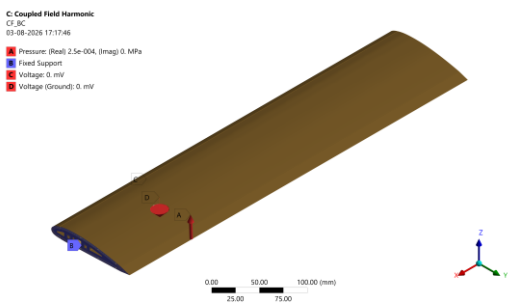


Fig. 7. Electrical and mechanical boundary conditions for coupled-field harmonic analysis in ANSYS Mechanical.

V. RESULTS AND DISCUSSION

A. Static Structural Analysis Results

Structural static simulations were carried out for the analysis of deformation, stiffness, and stress distribution of the wings due to aerodynamic loading. Table VIII shows static total deformations, equivalent Von Mises stress, and equivalent elastic strain for all five PZT materials.

TABLE VIII: STATIC STRUCTURAL ANALYSIS RESULTS

PZT Type	Max Deflection (mm)	von Mises Stress (MPa)	Elastic Strain (mm/mm)
PZT-4	0.14590	1.8991	2.9862×10^{-5}
PZT-5A	0.15830	2.0372	3.2522×10^{-5}
PZT-5H	0.15880	2.0324	3.3532×10^{-5}
PZT-8	0.14270	1.8490	2.8339×10^{-5}
PZT-26	0.13870	1.8923	2.7441×10^{-5}

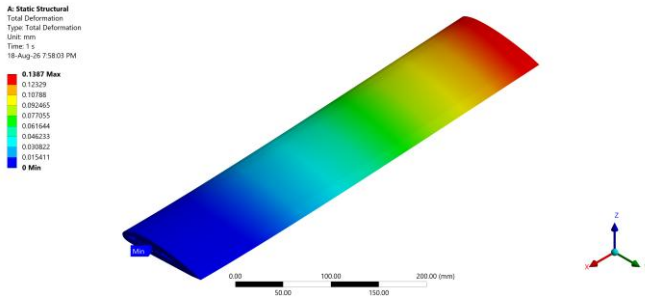


Fig. 8. Total deformation contour plot of the aircraft wing under steady aerodynamic pressure.

The PZT-26 exhibited the lowest peak total deformation (0.1387 mm) and equivalent elastic strain (2.7441×10^{-5} mm/mm). This implies maximum structural stiffness in relation to the test cases under the subjected load. The maximum Von-Mises stress values recorded for the different cases considered lie between 1.85 MPa and 2.04 MPa, values that are considerably below the 280 MPa value of yield strength stipulated for the alloy skin material.

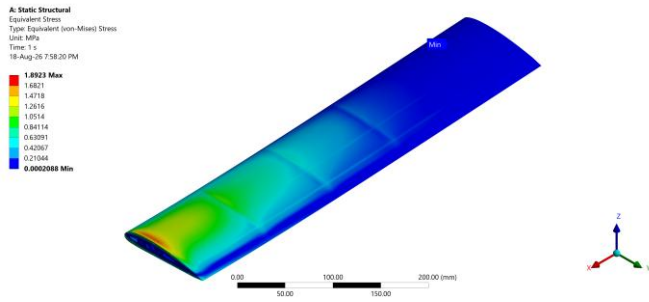


Fig. 9. Equivalent (von Mises) stress contour plot showing stress concentration near the fixed wing root.

B. Modal Analysis Results

Eigenvalues from free vibration analysis were extracted to obtain the basic natural frequency and modes. Table IX presents the first six natural frequencies for all materials investigated.

TABLE IX: NATURAL FREQUENCIES FOR THE FIRST SIX VIBRATION MODES

Vibration Mode	PZT-4	PZT-5A	PZT-5H	PZT-8	PZT-26
Mode 1	53.065	50.575	50.858	53.494	54.193
Mode 2	289.80	280.72	282.56	290.84	292.77
Mode 3	294.34	282.32	284.14	296.42	299.80
Mode 4	445.64	427.14	431.33	450.40	454.27
Mode 5	707.67	680.64	685.02	711.86	719.05
Mode 6	1145.2	1107.0	1114.5	1150.3	1160.3

The PZT-26 configuration had the maximum value of fundamental natural frequency at 54.193 Hz for Mode 1, which means it is stiffer dynamically compared to other configurations examined [5]. With higher value of fundamental frequency, it means that PZT-26 configuration might move the first structural resonance to higher frequency [2].

C. Coupled-Field Harmonic Analysis Results

Dynamic deflections, stresses, strains, and peak electrical voltages were derived from the coupled field harmonic frequency sweep through electromechanical coupling [1], [9]. The peak response values obtained close to the fundamental resonance are tabulated in Table X below.

TABLE X: PEAK RESULTS FROM COUPLED-FIELD HARMONIC ANALYSIS

PZT Type	Harmonic Deflect (mm)	Harmonic Stress (MPa)	Harmonic Strain (mm/mm)	Peak Voltage (mV)
PZT-4	2.9225e-2	5.7478	9.1112e-5	6111.1
PZT-5A	1.7716e-2	5.9043	6.0971e-5	5432.1
PZT-5H	1.6595e-2	5.6596	5.7432e-5	4185.0
PZT-8	3.6246e-2	6.4484	1.1134e-4	6275.1
PZT-26	3.7373e-2	5.9560	1.1308e-4	6117.6

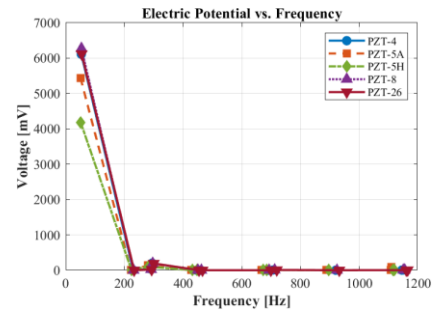


Fig. 10. Comparison of electrical potential frequency responses for the different piezoelectric materials.

PZT-8 had the maximum open-circuit electrical voltage output of 6275.1mV whereas PZT-26 gave an output of 6117.6mV, which is close to PZT-8's voltage value. Moreover, PZT-26 showed low static deflection and a high fundamental natural frequency [2], [7].

D. Comprehensive Performance Comparison

To identify the best possible candidate, the outputs for the various mechanical, dynamic, and electromechanical systems are all compared in Table XI [1], [7], [8].

TABLE XI: OVERALL PERFORMANCE COMPARISON OF THE PIEZOELECTRIC MATERIALS

Parameter	PZT-4	PZT-5A	PZT-5H	PZT-8	PZT-26
Static Deflection(mm)	0.1459	0.1583	0.1588	0.1427	0.138
Static Stress (MPa)	1.8991	2.0372	2.0324	1.849	1.8923
Static Strain(mm/mm)	2.986e-5	3.252e-5	3.353e-5	2.834e-5	2.744e-5
1st Nat Freq (Hz)	53.065	50.575	50.858	53.494	54.193
Harmonic Deflect (mm)	2.923e-2	1.772e-2	1.660e-2	3.625e-2	3.737e-2
Harmonic Stress (MPa)	5.7478	5.9043	5.6596	6.4484	5.9560
Harmonic Strain (mm/mm)	9.111e-5	6.097e-5	5.743e-5	1.113e-4	1.131e-4
Electrical Voltage (mV)	6111.1	5432.1	4185.0	6275.1	6117.6

Although PZT-8 recorded the peak voltage value of 6275.1 mV, PZT-26 recorded the minimum static deflection value of 0.1387 mm, maximum fundamental frequency of 54.193 Hz, and good voltage value of 6117.6 mV [2], [7]. As per the results obtained through structural and dynamic analysis, PZT-26 is regarded as the most favorable among all the configurations for the existing modeling, while PZT-8 generates maximum electric voltage.

VI. CONCLUSION AND FUTURE SCOPE

This study presented a comparative numerical analysis on the incorporation of five types of PZT ceramic material within a sandwich composite aircraft wing skin for structural health monitoring and energy harvesting purposes [1], [6]. CAD

modeling was done using CATIA V5 while finite element analysis (FEA) was performed using ANSYS Mechanical [2], [8].

Important findings drawn from this study are:

1) The maximum von Mises stress obtained under applied aerodynamic pressure of 250 Pa for the tested geometries varied in the range of 1.85-2.04 MPa, and was far lower than the 280 MPa yield strength of the aluminum alloy skin [4].

2) Best static stiffness in terms of the minimum deflection value of 0.1387 mm and highest fundamental natural frequency of 54.193 Hz were obtained when PZT-26 was used [5].

3) Harmonic coupled-field analysis predicted the production of electrical voltage by resonance; PZT-8 gave the maximum output voltage of 6275.1 mV, whereas PZT-26 gave 6117.6mV [7], [9].

4) PZT-26 can be regarded as the most promising material candidate based on the current modeling conditions owing to its structural stiffness and high fundamental natural frequency with a comparatively high voltage output [2].

Extensions are further wind-tunnel tests, fluid-structure interaction (FSI) aeroelasticity simulation, thickness and position optimization of PZT segments, and micro-circuits conditioning of the produced electrical energy [1], [6], [7].

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